KERALA DIPLOMA CURRICULUM REVISION 2026

Course 6371B | 6 Credits: 4 | Program Elective | Source-grounded

Sensors and Actuators

 To introduce the basic principles of different types of sensors, transducers and actuators.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 6371B is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 6371B.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Integrated Circuits Design & Fabrication
Course code	6371B
Course title	Sensors and Actuators
Semester	6 Credits: 4
Course category	Program Elective
Periods per week	4 (L:4 T:0 P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

13 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To introduce the basic principles of different types of sensors, transducers and actuators.
- To learn different transducers and their basic applications.
- To know the basic principles of micro actuators.

Course outcomes

CO	Official outcome	Study level
CO1	Explain the basics of transducers & sensors. 15 Applying Explain different types of inductive, capacitive	See official syllabus
CO2	14 Understanding and piezo electric transducers. Explain photo electric transducers, radiation	See official syllabus
CO3	detectors, ultrasonic detectors, Hall effect 14 Understanding transducers. Explain the concepts of actuators, electrical	See official syllabus
CO4	actuators and micro actuators. 15 Understanding	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 2
CO2	CO2 2
CO3	CO3 2
CO4	CO4 2

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Explain the basics of transducers & sensors	3	As specified
Module 2	transducers.	4	As specified
Module 3	detectors, Hall effect transducers.	3	As specified
Module 4	actuators	3	As specified

MODULE 1

Explain the basics of transducers & sensors

This module is presented from the official syllabus scope for Sensors and Actuators. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the basics of transducers & sensors
- Official duration: As specified
- Official topic entries captured: 3

- The full source excerpt is preserved below for coverage checking.

6 Understanding them. Classify different types of resistive

6 Understanding them. Classify different types of resistive is an official topic in Module 1 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding them. Classify different types of resistive using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding them. classify different types of resistive should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding them. Classify different types of resistive means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 6 Understanding them. Classify different types of resistive □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Understanding transducers Explain the applications of different types

6 Understanding transducers Explain the applications of different types is an official topic in Module 1 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding transducers Explain the applications of different types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding transducers explain the applications of different types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding transducers Explain the applications of different types means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1-□□ 6 Understanding transducers Explain the applications of different types □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric

3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric is an official topic in Module 1 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding of resistive transducers. introduction to transducers, resistive transducers: define transducer and sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. resistive transducers: basic principle of resistive transducers; working of linear potentiometer, loading effect; basic principle of strain gauge, different types, derivation of equation for gauge factor, construction and working of light dependent resistor (ldr), resistance temperature detector (rtd) and thermistor; displacement measurement using potentiometer. explain different types of inductive, capacitive and piezo electric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive

Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric [illegible] definition/meaning [illegible], steps, application [illegible]. Technical terms English-[illegible].

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the basics of transducers & sensors
               Explain transducers, sensors and classify
   M1.01                                                            6
Understanding
               them.
               Classify different types of resistive
   M1.02                                                            6
Understanding
               transducers
               Explain the applications of different types
   M1.03                                                            3
Understanding
               of resistive transducers.

 Contents:
 Introduction to Transducers, Resistive transducers: Define transducer and
Sensor;
 classification of transducers as active-passive, analog-digital, unidirectional
- bidirectional,
 mechanical – electrical, primary - secondary transducers.
 Resistive Transducers: Basic principle of resistive transducers; Working of
linear
 potentiometer, loading effect; Basic principle of strain gauge, different types,
derivation of
 equation for gauge factor, Construction and working of Light dependent resistor
(LDR),
 Resistance temperature detector (RTD) and Thermistor; Displacement measurement
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

transducers.

This module is presented from the official syllabus scope for Sensors and Actuators. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: transducers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of

transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of is an official topic in Module 2 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transducers and classification of inductive 3 understanding transducers. explain the construction and working of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types

LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types is an official topic in Module 2 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, lvdt and eddy current transducer and 4 understanding lvdt for pressure measurement. explain basic principle of capacitive transducers and working of different types should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric

4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric is an official topic in Module 2 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding of capacitive transducers and capacitive level gauge. explain working principle of piezo electric should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-[illegible] 4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic

transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic is an official topic in Module 2 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transducers and piezo electric pressure 3 understanding transducer. inductive, capacitive and piezo electric transducers: basic principle of inductive transducer; classification of inductive transducers; construction and working of linear variable differential transformer (lvdt); pressure measurement using lvdt; basic principle of eddy current transducers. illustrate basic principle of capacitive transducer; working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); level measurement using capacitive transducer. basic principle of piezo electric transducer; equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; equivalent circuit of piezo electric transducer; pressure measurement using piezo electric transducer. explain photo electric transducers, radiation detectors, ultrasonic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
transducers.
               Explain the basic principle of inductive
   M2.01       transducers and classification of inductive           3
Understanding
               transducers.
              Explain the construction and working of
  M2.02       LVDT and eddy current transducer and                  4
Understanding
              LVDT for pressure measurement.
              Explain basic principle of capacitive
              transducers and working of different types
  M2.03                                                             4
Understanding
              of capacitive transducers and capacitive
              level gauge.
              Explain working principle of piezo electric
  M2.04       transducers and piezo electric pressure               3
Understanding
              transducer.
              Series Test – I                                       1

Contents:
Inductive, Capacitive and Piezo electric transducers:
Basic principle of inductive transducer; Classification of inductive transducers;
Construction and working of Linear Variable Differential Transformer (LVDT);
pressure
measurement using LVDT; basic principle of Eddy current transducers.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

detectors, Hall effect transducers.

This module is presented from the official syllabus scope for Sensors and Actuators. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: detectors, Hall effect transducers.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic

photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic is an official topic in Module 3 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, photo emissive transducers (photo 5 understanding multiplier tube), solar cell and photo voltaic cell. explain the basic principle of ultrasonic should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a

diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

transducer and level measurement 4 Understanding application. Explain the construction and working

transducer and level measurement 4 Understanding application. Explain the construction and working is an official topic in Module 3 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify transducer and level measurement 4 Understanding application. Explain the construction and working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, transducer and level measurement 4 understanding application. explain the construction and working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What transducer and level measurement 4 Understanding application. Explain the construction and working means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-[illegible] transducer and level measurement 4 Understanding application. Explain the construction and working [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro

principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro is an official topic in Module 3 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.

- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, principle of hall effect transducer and 5 understanding current measurement application. photoelectric, ultrasonic and hall effect transducers: illustrate photoelectric transducers- working principle of photo conductive -photo diode, photo transistor- transducers. working principle of photo emissive -photo multiplier tube- transducers, working principle of solar cell and photo voltaic cell. working principle of ultra sonic transducer. construction and working principle of hall effect transducer. application of ultrasonic transducer for level measurement, hall effect transducers for current measurement. explain the concepts of actuators, electrical actuators and micro should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
detectors, Hall effect transducers.
              Explain photoelectric transducers- photo
              conductive (photo diode, photo transistor),
  M3.01       photo emissive transducers (photo                   5
Understanding
              multiplier tube), solar cell and photo voltaic
              cell.
              Explain the basic principle of Ultrasonic
  M3.02       transducer and level measurement                    4
Understanding
              application.
              Explain the construction and working
  M3.03       principle of Hall effect transducer and             5

Understanding
              current measurement application.
 Contents:
 Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric
transducers-
 Working principle of photo conductive -photo diode, photo transistor-
transducers. Working
 principle of photo emissive -photo multiplier tube- transducers, Working
principle of solar
 cell and photo voltaic cell.
 Working Principle of Ultra Sonic transducer. Construction and working principle
of Hall
 effect transducer. Application of Ultrasonic transducer for level measurement,
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

actuators

This module is presented from the official syllabus scope for Sensors and Actuators. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: actuators
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems.

Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems. is an official topic in Module 4 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the concept of actuators. 3 understanding explain different electrical actuating systems. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

6 Understanding Explain basic principles of micro actuators

6 Understanding Explain basic principles of micro actuators is an official topic in Module 4 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding Explain basic principles of micro actuators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding explain basic principles of micro actuators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding Explain basic principles of micro actuators means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-[illegible] 6 Understanding Explain basic principles of micro actuators [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, "A course in Electrical and Electronic measurement and R1 Instrumentation", Dhanpathrai and Co. R2 D.V.S. Murthy, " Transducers and Instrumentation", PHI. Pvt. Ltd. D.Patranabis, " Principles of Industrial Instrumentation", Tata McGraw-Hill. R3 .Education, J.B.Gupta, "Electrical and Electronic Measurements and instrumentation", R4 Kataria Pulishers. R5 Ranganathan S, " Transducer engineering ", Allied publishers. R6 Krishna Kant, "Computer-Based Industrial Control", PHI. R7 D. Patranabis, "Sensors and Transducers", PHI. R8 Massood Tabib and Azar, "Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures", First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, "Shape Memory Actuators", first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODIIO/TRANSDUCERS_ VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E

6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, "A course in Electrical and

Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. R5 Ranganathan S, “ Transducer engineering ”, Allied publishers. R6 Krishna Kant, “Computer-Based Industrial Control”, PHI. R7 D. Patranabis, “Sensors and Transducers”, PHI. R8 Massood Tabib and Azar, “Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures”, First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, “Shape Memory Actuators”, first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_ VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E is an official topic in Module 4 of Sensors and Actuators. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. R5 Ranganathan S, “ Transducer engineering ”, Allied publishers. R6 Krishna Kant, “Computer-Based Industrial Control”, PHI. R7 D. Patranabis, “Sensors and Transducers”, PHI. R8 Massood Tabib and Azar, “Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures”, First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, “Shape Memory Actuators”, first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODIlO/TRANSDUCERS_ VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 understanding & its different types. actuators: definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, applications. electrical actuating systems: solid-state switches, solenoids, dc motors, stepper motors, servo motors. micro actuators: actuation principle, shape memory effects-one way, two way and pseudo elasticity. types of micro actuators-electrostatic, magnetic, fluidic, inverse piezo effect, other principles. text / reference: t/r book title / author a.k. sawhney, “a course in electrical and electronic measurement and r1 instrumentation”, dhanpathrai and co. r2 d.v.s. murthy, “ transducers and instrumentation”, phi. pvt. ltd. d.patranabis, “ principles of industrial instrumentation”, tata mcgraw-hill. r3 .education, j.b.gupta, “electrical and electronic measurements and instrumentation”, r4 kataria pulishers. r5 ranganathan s, “ transducer engineering ”, allied publishers. r6 krishna kant, “computer-based industrial control”, phi. r7 d. patranabis, “sensors and transducers”, phi. r8 massood tabib and azar, “microactuators electrical, magnetic, thermal, optical, mechanical, chemical and smart structures”, first edition, kluwer academic publishers, springer, 1997. r9 manfred kohl, “shape memory actuators”, first edition, springer online resources: sl.no website link 1 https://www.youtube.com/watch?v=fyggy7lldrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/files/4-sensors_ch2.pdf 5 https://www.youtube.com/watch?v=xsbunuadlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=ancnrtjnlqm http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-odllo/transducers_ variable_resistive_capacitive_inductive_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=cckmercuxve 14 https://www.youtube.com/watch?v=7jq12uuqk6e should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. R5 Ranganathan S, “ Transducer engineering ”, Allied publishers. R6 Krishna Kant, “Computer-Based Industrial Control”, PHI. R7 D. Patranabis, “Sensors and Transducers”, PHI. R8 Massood Tabib and Azar, “Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures”, First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, “Shape Memory Actuators”, first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODlIO/TRANSDUCERS_

Aspect	What to prepare
	VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-�� 6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. R5 Ranganathan S, “ Transducer engineering ”, Allied publishers. R6 Krishna Kant, “Computer-Based Industrial Control”, PHI. R7 D. Patranabis, “Sensors and Transducers”, PHI. R8 Massood Tabib and Azar, “Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures”, First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, “Shape Memory Actuators”, first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODIlO/TRANSDUCERS_ VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
actuators
   M4.01       Explain the concept of actuators.                    3
Understanding

               Explain different electrical actuating
               systems.
   M4.02                                                            6
Understanding

               Explain basic principles of micro actuators
   M4.03                                                            6
Understanding
               & its different types.
               Series Test – II                                     1

Contents:
Actuators: Definition, types (basics only)-electric linear, electric rotary,
piezoelectric,
selection criteria of actuator, Applications.
Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper
motors,
Servo motors.
Micro Actuators: Actuation principle, shape memory effects-one way, two way and
pseudo
elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse
piezo effect,
other principles.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain 6 Understanding them. Classify different types of resistive. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding them. Classify different types of resistive*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding transducers Explain the applications of different types. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding transducers Explain the applications of different types*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer, loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding of resistive transducers. Introduction to Transducers, Resistive transducers: Define transducer and Sensor; classification of transducers as active-passive, analog-digital, unidirectional - bidirectional, mechanical – electrical, primary - secondary transducers. Resistive Transducers: Basic principle of resistive transducers; Working of linear potentiometer,* *loading effect; Basic principle of strain gauge, different types, derivation of equation for gauge factor, Construction and working of Light dependent resistor (LDR), Resistance temperature detector (RTD) and Thermistor; Displacement measurement using potentiometer. Explain different types of inductive, capacitive and piezo electric*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Module 2

Define or explain transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of. (3 marks)

Answer: Begin with the standard definition or identity of *transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types. (3 marks)

Answer: Begin with the standard definition or identity of *LVDT and eddy current transducer and 4 Understanding LVDT for pressure measurement. Explain basic principle of capacitive transducers and working of different types*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□□ application □□□□ □□□□□□□□□□ □□□□□□□.

Define or explain 4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding of capacitive transducers and capacitive level gauge. Explain working principle of piezo electric*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic. (3 marks)

Answer: Begin with the standard definition or identity of *transducers and piezo electric pressure 3 Understanding transducer. Inductive, Capacitive and Piezo electric transducers: Basic principle of inductive transducer; Classification of inductive transducers; Construction and working of Linear Variable Differential Transformer (LVDT); pressure measurement using LVDT; basic principle of Eddy current transducers. Illustrate basic principle of capacitive transducer; Working of different types of capacitive transducers (variable area, variable distance, variable dielectric type); Level measurement using capacitive transducer. Basic principle of piezo electric transducer; Equation for voltage sensitivity and charge sensitivity of the piezo electric transducer; Equivalent circuit of piezo electric transducer; Pressure measurement using piezo electric transducer. Explain photo electric transducers, radiation detectors, ultrasonic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic. (3 marks)

Answer: Begin with the standard definition or identity of *photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain transducer and level measurement 4 Understanding application. Explain the construction and working. (3 marks)

Answer: Begin with the standard definition or identity of *transducer and level measurement 4 Understanding application. Explain the construction and working*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of

Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro. (3 marks)

Answer: Begin with the standard definition or identity of *principle of Hall effect transducer and 5 Understanding current measurement application. Photoelectric, Ultrasonic and Hall Effect Transducers: Illustrate photoelectric transducers- Working principle of photo conductive -photo diode, photo transistor- transducers. Working principle of photo emissive -photo multiplier tube- transducers, Working principle of solar cell and photo voltaic cell. Working Principle of Ultra Sonic transducer. Construction and working principle of Hall effect transducer. Application of Ultrasonic transducer for level measurement, Hall Effect Transducers for current measurement. Explain the concepts of actuators, electrical actuators and micro*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems.. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding Explain basic principles of micro actuators. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding Explain basic principles of micro actuators*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, “A course in Electrical and Electronic measurement and R1 Instrumentation”, Dhanpathrai and Co. R2 D.V.S. Murthy, “ Transducers and Instrumentation”, PHI. Pvt. Ltd. D.Patranabis, “ Principles of Industrial Instrumentation”, Tata McGraw-Hill. R3 .Education, J.B.Gupta, “Electrical and Electronic Measurements and instrumentation”, R4 Kataria Pulishers. R5 Ranganathan S, “ Transducer engineering ”, Allied publishers. R6 Krishna Kant, “Computer-Based Industrial Control”, PHI. R7 D. Patranabis, “Sensors and Transducers”, PHI. R8 Massood Tabib and Azar, “Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures”, First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, “Shape Memory Actuators”, first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- 8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_ VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E. (3 marks)

Answer: Begin with the standard definition or identity of *6 Understanding & its different types. Actuators: Definition, types (basics only)-electric linear, electric rotary, piezoelectric, selection criteria of actuator, Applications. Electrical Actuating Systems: Solid-state switches, Solenoids, DC motors, Stepper motors, Servo motors. Micro Actuators: Actuation principle, shape memory effects-one way, two way and pseudo elasticity. Types of micro actuators- Electrostatic, Magnetic, Fluidic, Inverse piezo effect, other principles. Text / Reference: T/R Book Title / Author A.K. Sawhney, "A course in Electrical and Electronic measurement and R1 Instrumentation", Dhanpathrai and Co. R2 D.V.S. Murthy, " Transducers and Instrumentation", PHI. Pvt. Ltd. D.Patranabis, " Principles of Industrial Instrumentation", Tata McGraw-Hill. R3 .Education, J.B.Gupta, "Electrical and Electronic Measurements and instrumentation", R4 Kataria Pulishers. R5 Ranganathan S, " Transducer engineering ", Allied publishers. R6 Krishna Kant, "Computer-Based Industrial Control", PHI. R7 D. Patranabis, "Sensors and Transducers", PHI. R8 Massood Tabib and Azar, "Microactuators Electrical, Magnetic, thermal, optical, mechanical, chemical and smart structures", First edition, Kluwer academic publishers, Springer, 1997. R9 Manfred Kohl, "Shape Memory Actuators", first edition, Springer Online Resources: Sl.No Website Link 1 https://www.youtube.com/watch?v=FYggY7LLDrw 2 https://www.slideshare.net/manash234/classification-of-transducers 3 https://www.electronics-tutorials.ws/io/io_1.html 4 https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf 5 https://www.youtube.com/watch?v=XSbUnUADlno 6 http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html 7 https://www.youtube.com/watch?v=anCnrtjNLQM http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt-8 tutorial.pdf 9 https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers- 10 work/ https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen 11 tation 12 https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ 13 https://www.youtube.com/watch?v=CckMErCuXvE 14 https://www.youtube.com/watch?v=7jQ12uUqK6E*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **6 Understanding them. Classify different types of resistive**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **transducers and classification of inductive 3 Understanding transducers. Explain the construction and working of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **photo emissive transducers (photo 5 Understanding multiplier tube), solar cell and photo voltaic cell. Explain the basic principle of Ultrasonic**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain the concept of actuators. 3 Understanding Explain different electrical actuating systems.**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.

Term	Meaning in this lesson
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link https://www.youtube.com/watch?v=FYggY7LLDrw
- https://www.slideshare.net/manash234/classification-of-transducers
- https://www.electronics-tutorials.ws/io/io_1.html
- https://www.egr.msu.edu/classes/ece445/mason/Files/4-Sensors_ch2.pdf https://www.youtube.com/watch?v=XSbUnUADlno
- http://www.kyowa-ei.com/eng/technical/strain_gages/principles.html https://www.youtube.com/watch?v=anCnrtjNLQM
- http://www.te.com/content/dam/te-com/documents/sensors/global/lvdt- tutorial.pdf
- https://www.electronics-tutorials.ws/electromagnetism/hall-effect.html
- https://www.brighthubengineering.com/hvac/53147-how-capacitive-transducers-
- https://www.powershow.com/view4/556e94-ODllO/TRANSDUCERS_
- VARIABLE_RESISTIVE_CAPACITIVE_INDUCTIVE_powerpoint_ppt_presen
- https://www.ctscorp.com/resource-center/tutorials/piezo-basics/ https://www.youtube.com/watch?v=CckMErCuXvE https://www.youtube.com/watch?v=7jQ12uUqK6E

IN-PAGE SEARCH

Search this lesson